

5(a)	At $V = 1.5\text{V}$, $I = 0.2\text{ A} \Rightarrow$ resistance $R = \frac{V}{I} = \frac{1.5}{0.2} = 7.5\ \Omega$ At $V = 3.0\text{V}$, $I = 0.3\text{ A} \Rightarrow$ resistance $R = \frac{V}{I} = \frac{3.0}{0.3} = 10\ \Omega$ Change in resistance = $10 - 7.5 = 2.5\ \Omega$
(b)(i)	From the graph, when current $I = 0.36\text{ A}$, the potential difference across the lamp is 4.5 V. Hence terminal potential difference = $4.5 + 0.36 \times 5.0 = 6.3\text{ V}$.

	Current flowing in $14\ \Omega$ resistor $= \frac{6.3}{14} = 0.45\text{ A}$
(b)(ii)	Using $E = I(R+r)$ $7.5 = IR + Ir$ $= 6.3 + (0.45 + 0.36)r$ $r = 1.48\ \Omega$
6(a)	Faraday's law of electromagnetic induction states that the induced e m f in a coil is